

LANCANG, China

WMO: 569540

Lat: 22.567N Lon: 99.933E Elev: 3458 StdP: 12.95 Time Zone: 8.00 (E08) Period: 94-19 WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 42.1	(c) 44.0	(d) 35.2	(e) 34.1	(f) 73.4	(g) 38.1	(h) 38.3	(i) 67.2	(j) 13.6	(k) 73.8	(l) 11.8	(m) 73.4	(n) 0.5	(o) 160	(p) 0.233

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	12.7	90.6	66.7	88.7	66.9	86.9	67.4	73.8	82.0	73.3	81.1	72.7	80.0	7.0	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.8	134.0	76.1	71.2	130.9	75.5	70.8	129.3	75.2	40.1	82.5	39.5	81.6	39.0	80.5	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.9	9.9	8.1	37.9	93.7	3.6	1.9	35.3	95.1	33.3	96.2	31.3	97.3	28.7	98.6		
			37.4	75.1	3.5	0.8	34.8	75.6	32.7	76.1	30.8	76.5	28.2	77.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.3	60.3	62.7	67.7	72.9	75.6	76.4	75.5	75.9	75.1	72.3	66.6	61.5
	DBStd	6.53	2.59	2.77	2.84	2.92	3.23	2.28	2.44	2.28	2.74	2.90	3.39	3.60
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	370	147	72	9	1	1	0	0	0	0	1	21	118
	CDD50	7396	319	357	550	686	794	793	792	804	754	692	499	356
	CDD65	2292	1	9	94	237	330	343	327	339	305	228	70	9
	CDH74	14785	109	489	1374	2339	2537	1885	1453	1634	1511	1022	351	81
	CDH80	4094	1	62	420	946	976	469	292	364	353	189	21	1
(14) Wind	WSAvg	2.5	2.4	2.9	3.2	3.2	2.8	2.5	2.3	2.3	2.4	2.2	2.1	2.0
(15) Precipitation	PrecAvg	64.5	0.7	0.6	0.6	1.7	6.3	10.1	13.5	12.4	7.9	6.0	3.3	1.1
	PrecMax	82.6	3.4	1.6	3.3	4.6	13.2	18.3	19.8	20.6	17.6	15.0	10.1	5.1
	PrecMin	53.3	0.0	0.0	0.0	0.2	1.4	5.2	6.9	5.3	3.2	0.7	0.2	0.1
	PrecStd	8.0	0.9	0.5	0.7	1.1	3.3	3.3	3.1	4.3	3.1	3.1	2.6	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.0	84.4	88.7	92.5	93.7	90.8	88.3	89.0	88.2	86.2	82.6	78.7
		MCWB	57.7	59.0	60.9	64.4	66.8	70.7	73.0	72.2	71.5	69.5	66.1	62.6
	2%	DB	76.9	81.9	86.6	90.3	90.9	87.5	85.5	86.2	86.1	84.1	80.1	76.2
		MCWB	57.2	58.5	60.5	64.0	67.5	71.3	72.8	72.3	71.3	68.6	64.6	60.7
	5%	DB	75.1	79.9	84.8	88.4	88.6	85.0	83.3	84.0	84.1	82.3	78.1	74.4
		MCWB	57.2	57.8	60.1	63.7	67.6	71.6	72.4	72.3	70.8	68.2	63.8	59.9
	10%	DB	72.7	77.3	82.4	86.1	85.8	82.5	81.0	81.7	81.9	80.0	75.9	72.0
		MCWB	56.7	57.0	59.6	63.7	68.1	71.6	72.0	72.0	70.7	68.1	63.2	59.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.4	62.4	65.5	68.9	72.6	74.2	74.7	74.7	74.0	72.3	69.0	66.0
		MCDB	70.9	77.6	77.5	81.3	80.8	82.9	83.4	83.4	82.6	80.5	76.4	72.9
	2%	WB	60.6	60.4	63.8	67.4	71.6	73.5	73.7	73.7	73.0	71.0	67.5	64.1
		MCDB	70.1	76.4	76.3	80.2	80.8	81.3	81.7	82.0	81.0	79.3	74.7	71.3
	5%	WB	59.1	59.1	62.6	66.4	70.9	72.8	72.9	72.9	72.4	70.2	66.5	62.1
		MCDB	69.9	74.6	77.0	79.8	79.7	80.1	80.1	80.6	80.0	78.1	73.4	69.5
	10%	WB	57.7	58.0	61.4	65.4	70.1	72.2	72.4	72.4	71.7	69.4	65.3	60.5
		MCDB	68.8	73.3	76.5	79.5	78.6	78.9	78.8	79.3	78.8	76.7	71.9	68.4
(35) Mean Daily Temperature Range	5% DB	MDBR	27.8	31.2	30.2	26.6	19.1	12.7	10.8	12.3	14.2	16.3	20.0	22.7
		MCDBR	31.4	35.1	34.1	30.5	25.3	17.4	15.1	16.2	17.8	20.0	22.5	25.5
		MCWBR	13.3	12.8	10.3	7.9	6.0	4.3	4.1	4.5	5.0	6.3	8.4	10.9
	5% WB	MCDBR	25.6	30.1	26.9	24.0	16.5	13.3	12.5	13.7	14.5	15.6	16.9	20.3
		MCWBR	11.5	11.7	9.1	7.1	4.7	3.9	3.9	4.4	4.6	5.3	6.6	9.2
(40) Clear-Sky Solar Irradiance	taub		0.357	0.417	0.673	0.652	0.474	0.426	0.422	0.455	0.454	0.448	0.382	0.366
	taud		2.351	2.179	1.546	1.650	2.198	2.386	2.385	2.264	2.242	2.220	2.346	2.355
	Ebn at Noon		287	275	213	222	264	274	275	267	266	262	277	280
	Edh at Noon		36	45	89	82	47	39	39	44	44	43	36	35
(44) All-Sky Solar Radiation	RadAvg		1366	1678	1722	1793	1625	1400	1264	1358	1382	1295	1306	1196
(45)	RadStd		88	108	102	128	160	117	96	99	90	117	99	95

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.61	+0.92	+1.15	N/A	N/A	N/A	N/A	N/A	+252	+229
Regional (6 neighbors)	+0.58	N/A	N/A	+0.90	N/A	N/A	N/A	-79	+259	+189